

# A Self-Consistent Spatially Resolved Cluster Dynamics Model for Irradiated Zirconium

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Multiscale Materials Solutions — Deliverable D1/M2

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## Three things

- A **self-consistent** cluster-dynamics model: every bias factor is generated from the *same* diffusion tensors that transport the mobile species.
- A **continuum** → **discrete** conversion that preserves stored defects exactly and loop number to the integer draw.
- A **two-time-scale integration** that makes  $2.1 \times 10^8$  stiff equations tractable.

## Demonstrated on

- A 500 nm hexagonal  $\alpha$ -Zr single crystal, 800 nm tall.
- $T = 573$  K,  $G = 10^{-7}$  dpa s<sup>-1</sup>, zero applied stress.
- 0 → 40 dpa in 13 dose intervals, 90 663 nodes.
- Calibrated against **78 TEM measurements** across twelve temperatures and four decades of dose.

The model is self-consistent in two senses — in the bias factors, and in the continuum/discrete transfer — and both are load-bearing for what follows.

|                      | D1/M1 (SRCD)                      | D1/M2 (this work)                      |
|----------------------|-----------------------------------|----------------------------------------|
| Immobile families    | 4 (aligned / non-aligned)         | <b>9</b> crystallographic              |
| Moments carried      | density, content                  | + <b>second moment</b> $q$             |
| Capture efficiencies | phenomenological $Z = 1 + \delta$ | <b>Woo</b> , from the diffusion tensor |
| Size distribution    | not resolved                      | <b>log-normal closure</b>              |
| Grain boundary       | Dirichlet, mobile only            | + <b>size-gated loop removal</b>       |
| Discrete loops       | visualization only                | <b>conserving conversion</b>           |
| Calibration          | —                                 | <b>MAP against 78 measurements</b>     |

The single structural change is that the anisotropy now enters *once*, through the diffusion tensors, and reaches the loops through the capture efficiencies rather than through an independent set of fitted biases.

# The nine immobile families

| Family                | Habit                         | Character    | Burgers vector                          |
|-----------------------|-------------------------------|--------------|-----------------------------------------|
| $a_1, a_2, a_3$       | prismatic $\{10\bar{1}0\}$    | interstitial | $\frac{1}{3}\langle 11\bar{2}0 \rangle$ |
| $a_1^v, a_2^v, a_3^v$ | prismatic $\{10\bar{1}0\}$    | vacancy      | $\frac{1}{3}\langle 11\bar{2}0 \rangle$ |
| $c_0$                 | basal, stacking-fault pyramid | vacancy      | — (no line)                             |
| $c_f$                 | basal (0001), faulted         | vacancy      | $\frac{1}{2}[0001]$                     |
| $c_p$                 | basal (0001), perfect         | vacancy      | $\frac{1}{2}[0001]$                     |

$|b_{\langle a \rangle}| = a = 3.23 \text{ \AA}$  and  $|b_{\langle c \rangle}| = c/2 = 2.575 \text{ \AA}$  with the *physical* axial ratio  $c/a = 1.5944$ , not the ideal  $\sqrt{8/3}$ . The same four numbers are used by the 0-D model, the continuum solver and the discrete conversion — a mismatch there was a factor of two in the  $\langle c \rangle$  Burgers vector.

### Mobile — $v, i, 2i, 3i$

- Four concentration fields  $\mathbf{c}_M(\mathbf{x})$ .
- **Spatially coupled** by diffusion.
- **No time derivative**: a steady reaction–diffusion problem.
- Dirichlet at grain boundaries.
- $4 \times 90\,663 = 362\,652$  unknowns.

### Immobile — nine loop families

- Three moments each:  $n_{sL}^k, c_{sL}^k, q_{sL}^k$ .
- **Pointwise**: no neighbor's state appears.
- Stiff ODEs in dose.
- $27 \times 90\,663 = 2\,447\,901$  unknowns.

That asymmetry — one block coupled and algebraic, the other pointwise and stiff — is what the operator split of §4 exploits, and it is the reason the calculation is affordable at all.

# The mobile master equation

$$\mathbf{0} = \nabla \cdot (\mathbf{D}_{\text{ten}} \nabla \mathbf{c}_M) + \mathbf{P}_M + \frac{1}{2}(\mathbf{I} \otimes \mathbf{c}_M^T) \mathbf{R}_2 \mathbf{c}_M + (\mathbf{E} - \mathbf{D}) \mathbf{c}_M + \mathbf{S}_{vL}$$

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|                                                                            |                                                                     |
|----------------------------------------------------------------------------|---------------------------------------------------------------------|
| $\nabla \cdot (\mathbf{D}_{\text{ten}} \nabla \mathbf{c}_M)$               | anisotropic transport of the four mobile species                    |
| $\mathbf{P}_M$                                                             | cascade production (dose rate $\times$ survival $\times$ partition) |
| $\frac{1}{2}(\mathbf{I} \otimes \mathbf{c}_M^T) \mathbf{R}_2 \mathbf{c}_M$ | second-order reactions: recombination and clustering                |
| $\mathbf{E} \mathbf{c}_M$                                                  | thermal dissociation of mobile clusters                             |
| $-\mathbf{D} \mathbf{c}_M$                                                 | absorption at immobile sinks: network + loops                       |
| $\mathbf{S}_{vL}$                                                          | vacancies returned by peripheral emission from loops                |

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Steady, not quasi-steady:  $\partial_t \mathbf{c}_M$  is absent by construction, which is what allows the coupled problem to be solved a handful of times over a march rather than at every step.

Absorption: where the loops act back on the mobile species

$$\mathbf{D} = \text{diag}\{\boldsymbol{\omega} \odot \mathbf{C}_s\} + \mathbf{D}_d \mathbf{K}_{\mathcal{L}}, \quad C_s^v = \frac{a^2}{z_c} \rho_N, \quad C_s^i = \frac{a^2}{z_c} Z_N \rho_N$$

With  $\omega^m = z_c \nu^m e^{-E_{\text{mig}}^m / k_B T}$  and  $\bar{D}^m = (a^2 / z_c) \omega^m$ , the first term is simply

$$\boldsymbol{\omega} \odot \mathbf{C}_s = (\bar{D}^v \rho_N, Z_N \bar{D}^i \rho_N, Z_N \bar{D}^{2i} \rho_N, Z_N \bar{D}^{3i} \rho_N)$$

$a^2$  and  $z_c$  cancel: the network term is the classical  $\bar{D}^m k_N^2$ . Splitting it into a frequency and a sink concentration is 0-D bookkeeping, not a second physical quantity.

The loops enter the *second* term only, through  $k_m^2 = \sum_{(s,k)} Z_{sk,m} \rho_{sL}^k + \dots$  — they must not also appear in  $\mathbf{C}_s$ , or their absorption would be counted once as a network sink and once as loop growth.

# Anisotropy enters once, through one number per species

Migration energies are split at fixed  $\bar{D}$ :

$$E_m^{(11,22)} = E_m^{\text{eff}} + 2k_B T \ln p^m$$

$$E_m^{(33)} = E_m^{\text{eff}} - 4k_B T \ln p^m$$

so a change in  $p^m$  alters *directionality*, never overall mobility.

The same  $p^m$  builds the Woo capture efficiencies:

$$Z_{\text{basal}} = Z_m^0 p^m, \quad Z_{\text{prism}} = Z_m^0 \frac{p^m + p_m^{-2}}{2}$$

This is the point of the self-consistency: because every bias follows from the diffusion tensors, the anisotropy *cannot* absorb a shortfall by refitting a bias. It is an input, and what is left over has to appear somewhere that can be named.

Fixed from transport, not from loop counts

$$p^v = 1.02, \quad p^i = 0.70$$

implying an interstitial migration anisotropy of **0.106 eV** at 573 K — a falsifiable prediction, open to atomistic test.

## Simultaneous $\langle a \rangle$ and $\langle c \rangle$ growth

Both growth conditions reduce to bounds on a single number — the arrival ratio

$$A/B = \overline{D}^v c_v / \sum_m \overline{D}^m c_m |m|$$

and the window between them has width  $g(p^v)/g(p^i)$  with  $g(p) = 2/(1 + p^{-3})$ , strictly increasing. Hence

Co-growth of both families is possible **if and only if**  $p^i < p^v$ .

The calibrated pair  $p^i = 0.70 < p^v = 1.02$  satisfies it with margin.

Two limits worth stating: the criterion governs the *absorption flux* only — cascade nucleation is unaffected by anisotropy — and an isotropic calculation is another parameter point, not a zero of the criterion, so a ratio taken against one cannot test a statement about signs.

## The immobile block: three moments per family

Restricting the master equation to indices that do not diffuse leaves a birth–death equation on the integer size  $m$ . Three successive reductions:

1. Treat  $m$  as continuous, truncate Kramers–Moyal at second order  $\Rightarrow$  Fokker–Planck in  $m$ .
2. Multiply by  $m^j$ , integrate  $\Rightarrow$  an *exact* moment hierarchy; retain  $j = 0, 1, 2$ .
3. Close it with a log-normal  $f_k(m)$ .

$$n_{sL}^k = \int f_k dm, \quad c_{sL}^k = \int m f_k dm, \quad q_{sL}^k = \int m^2 f_k dm$$
$$\bar{m}_{sL}^k = \frac{c_{sL}^k}{n_{sL}^k}, \quad \Delta_k \equiv \frac{q_{sL}^k n_{sL}^k}{(c_{sL}^k)^2} = e^{s_k^2} \geq 1$$

$\Delta_k \geq 1$  is Cauchy–Schwarz on a non-negative measure. It is not a tolerance:  $\Delta < 1$  is a defect in the equations, and it is the sharpest available check that every channel is closed consistently.

## The second moment earns its place

The Fokker–Planck contribution to  $\dot{q}$  is

$$\dot{q}_{sL}^k \Big|_{\text{FP}} = 2\Pi_{sL}^k \left[ \underbrace{\sum_k^{\text{net}} \bar{m}_{sL}^k \Delta_k^{3/8}}_{\text{drift}} + \underbrace{\frac{1}{2} \sum_k^{\text{gro}} \Delta_k^{-1/8}}_{\text{spreading}} \right]$$

At  $\Sigma_k^{\text{net}} = 0$  the drift vanishes but  $\dot{q}_{sL}^k = \Pi_{sL}^k \Sigma_k^{\text{gro}} \Delta_k^{-1/8} > 0$ : the distribution **keeps broadening at fixed mean size and fixed number**. No pair of lower moments can express that.

The closure reaches the rest of the model in exactly one place — as  $\Delta_k^{-1/8}$  on the sink prefactor — and reaches the discrete conversion as the distribution loop sizes are drawn from.

Drift carries  $\Delta^{3/8}$  and spreading  $\Delta^{-1/8}$ : a family that has already broadened broadens *faster*. The dispersion is self-amplifying through the drift, which is a real feature and, at one node in the calculation, a visible one.

## Loss channels, and the sign that is easy to get wrong

Coalescence, by an Avrami overlap on the family's own size:

$$\varphi_{LL} = 1 - e^{-\kappa_{LL} \frac{4}{3} \pi R_{LL}^3 N}, \quad \varphi_{LN} = 1 - e^{-\kappa_{LN} \pi R_{LN}^2 \rho_N}$$
$$\Delta N = \nu_{LL} \varphi_{LL} n + \Delta N_{LN}, \quad \Delta Q = \Delta_k (\bar{m}^2 \Delta N - 2\bar{m} \Delta C)$$

### A coalescing loop does not leave the family — it MERGES

Its defects stay; only  $\Delta C_{LN}$  is lost. Fewer loops holding the same content are *larger* loops, so like-loop coalescence **raises**  $q$ . Debiting  $\Delta N \langle m^2 \rangle$  — the natural-looking reading — inverts the sign on the largest single term in the equation.

Measured consequence of getting it wrong:  $q$  crossed zero at 1 dpa and was negative on 40% of the  $c_p$  nodes at 10 dpa. Coalescence carries  $\sim 98\%$  of the loop-number loss, so this is not a small term.

# The distribution leaks at both ends

|                     | Small end (vanishing)                             | Large end (coalescence)        |
|---------------------|---------------------------------------------------|--------------------------------|
| Removes             | loops below $m_{\text{van}}$                      | the biggest loops              |
| Weight              | $1 - \Phi^{(j)}$ , <i>below</i> the mean          | $\Phi^{(j)}$ , <i>above</i> it |
| Effect on $\bar{m}$ | <b>raises</b> it                                  | lowers it                      |
| Realizability       | automatic — what remains is a genuine sub-measure |                                |

$$\Xi_j^k = [1 - \Phi_k^{(j)}(m_{\text{van}})] \times \{n_{sL}^k, c_{sL}^k, q_{sL}^k\}, \quad \Phi_k^{(j)}(m_\theta) = \frac{1}{2} \operatorname{erfc} \left( \frac{\ln m_\theta - \mu_k - j s_k^2}{s_k \sqrt{2}} \right)$$

Because  $\Phi^{(2)} > \Phi^{(1)} > \Phi^{(0)}$ , the complements reverse that ordering: what the small-end leak removes is *smaller* than the family mean, so **the density falls while the surviving mean size rises**.

That is the one trade coalescence cannot make. It was the first channel found that moves density and diameter the way the measurements want them moved — fewer *and* larger.

## Mobile — Dirichlet

$$c^v = e^{-E_v^f/k_B T}, \quad c^i = c^{2i} = c^{3i} = 0$$

on  $\partial\Omega_D$ . Thermal equilibrium is imposed; the layer that develops is a result, not an input.

## Loops — size-gated removal

A loop of radius  $R^k = \lambda_k \sqrt{m}$  at distance  $x$  touches the face when

$$m \geq m_{gb}(x) = (x/\lambda_k)^2$$

so the absorbed fraction of each moment is  $\Theta_j^k = \Phi_k^{(j)}(m_{gb}) \times \{n, c, q\}$ .

A loop cannot be given a boundary condition — it is untransported and has no flux to set. What it has is a *size*, and the family already carries the distribution of sizes needed to say what fraction of it reaches the wall. The channel costs no new parameter beyond a rate  $\nu_{gb}$ .

# The operator split, and why it is legitimate

Lie–Trotter, alternating over each substep:

1. **Fast solve** — the steady mobile field  $\mathbf{c}_M^*(x)$  for the immobile state currently held.
2. **Slow march** — the immobile ODEs integrated at every point with the mobile species *frozen*.

## The decoupling is by construction, not by approximation

The only thing coupling neighboring points is diffusion of the mobile species. Freeze the mobile field and the spatial coupling is *gone*: a point's immobile ODEs depend on its own loop populations and the four numbers pinned into  $\mathbf{c}_M$ .

Splitting error is first order in the dose spanned between fast solves — so the fast-solve cadence, not the substep count, is the accuracy dial.

The bill is paid in splitting error and nowhere else. The coupled problem has not been approximated away; it has been moved to where it is affordable.

## What the split buys, in numbers

| flops for one implicit factorization             |                                        |
|--------------------------------------------------|----------------------------------------|
| Split: 73 518 independent $36 \times 36$ systems | $3.4 \times 10^9$                      |
| Coupled: one 3 263 868-dimensional system        | $3.5 \times 10^{19}$                   |
| <b>Ratio</b>                                     | <b><math>1.0 \times 10^{10}</math></b> |

### Measured, 500 nm case

|                                  |               |
|----------------------------------|---------------|
| march total                      | 1 h 10 m      |
| one fast solve ( $\times 26$ )   | 1 m 54 s      |
| one slow substep ( $\times 78$ ) | 15.3 s        |
| dedup factor                     | $\times 1.23$ |

### Why frequency matters

The fast solve has *no time derivative*, so the coupled problem is solved **26 times over the march** rather than at every timestep. The counterfactual is  $\sim 25$  h for the 4-species mobile block alone.

Sparsity softens the coupled figure but does not rescue it: a 3-D FE Jacobian still factorizes at  $O(N^2)$  with fill growing as  $O(N^{4/3})$ .

## Continuum → discrete: what must be preserved

| Invariant across a transfer | Requirement | Achieved                   |
|-----------------------------|-------------|----------------------------|
| Stored defects              | exact       | to $10^{-16}$ (rescaled)   |
| Loop number                 | exact       | to the integer draw        |
| Size dispersion             | preserved   | to the sampling truncation |
| Sink strength               | continuous  | 0.97–1.04 at every dose    |

$$\lambda_{sj}^k = \frac{n_{sL}^k(\mathbf{x}_j)}{\Omega} V_j, \quad \Lambda_s^k = \sum_j \lambda_{sj}^k, \quad N_{\text{draw}} = \text{round}(\Lambda_s^k)$$

A calculation may therefore be run as continuum, as discrete, or as a hybrid, with the choice made on grounds of **dose and cost** rather than of physics.

Rounding rather than drawing a Poisson total is deliberate:  $\text{Var} = \Lambda$  there, and a dose sweep whose count fluctuates by  $\sqrt{\Lambda}$  between snapshots is dominated by counting noise.

## Nodal volumes, and what a sparse cell means

$V_j$  is estimated by **Monte Carlo**: points drawn uniformly in the bounding box, rejected against the convex hull, each survivor assigned to its nearest node.

$$V_j = \frac{M_j}{M} V_\Omega, \quad M = 4 \times 10^6, \quad \text{error} \sim M^{-1/2}$$

### The one condition

The cells must tile the crystal exactly. They do: every accepted sample is charged to exactly one node, so  $\sum_j V_j = V_\Omega$  identically, however coarse the sampling. Error moves volume *between* cells; it cannot change their sum.

### Two things it is not

Not the FEM mesh — the cells are built on the concentration trial-function nodes and share only the point set. And a node with *zero* volume is a correct answer, not a failure:  $\sim 12\%$  receive none.

No cell expects a whole loop:  $\lambda \sim 10^{-2}$  typically. The draw is taken once for the *family* and the nodes sampled from the multinomial — rounding node by node would return zero everywhere and convert nothing.

## Calibration: what is fitted and what is not

| Class           | Examples                               | Fitted?              |
|-----------------|----------------------------------------|----------------------|
| Model controls  | which families, which channels         | no — selects physics |
| Crystallography | $a$ , $c$ , $\Omega$ , Burgers vectors | no — measured        |
| Transport       | $p^v = 1.02$ , $p^i = 0.70$            | <b>no — pinned</b>   |
| Rate parameters | capture prefactors, coalescence        | <b>yes</b>           |

**Targets:** 78 TEM measurements — 73  $\langle a \rangle$  and 5  $\langle c \rangle$ , twelve temperatures, four decades of dose.

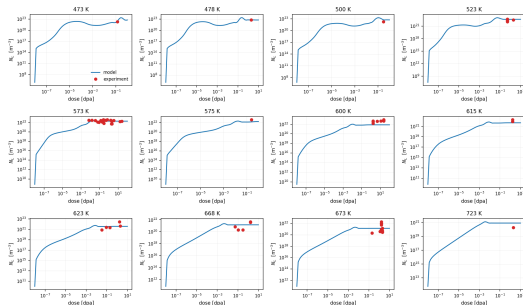
**Estimator:** maximum *a posteriori* under stated priors, on  $\log_{10}(\text{model/experiment})$ , symmetric under inversion of the ratio.

**Search:** staged bounded Nelder–Mead, polished jointly. One objective evaluation is 0.2–0.5 s, which is what makes a refit tractable.

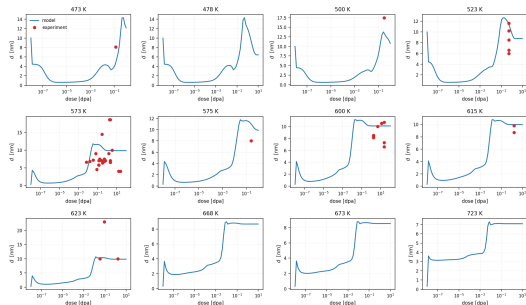
The 0-D estimator and the 3-D slow step are the *same binary with the same switches*, differing only in whether the mobile species are frozen — so a parameter set fitted in 0-D is one the 3-D code runs term for term.

# Prismatic $\langle a \rangle$ loops against experiment

(a) prismatic (interstitial) – density

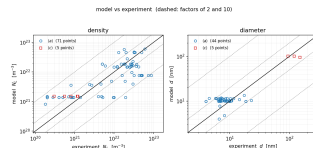
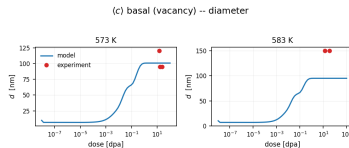
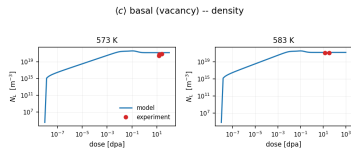


(a) prismatic (interstitial) – diameter



Number density (left) and diameter (right) by temperature. Density  $0.88\times$  the measurement over 71 points and diameter  $1.21\times$  over 44, across twelve temperatures and four decades of dose.

# Basal $\langle c \rangle$ loops, and the parity plot



On the *fixed* anisotropy the calibration reproduces the basal loop diameter without it being adjusted — **100.7 nm against 95 nm measured**. What it does not reach is the basal density, which remains  $2.1\times$  the measurement.

# What the calibration does not achieve

## A prismatic temperature trend

The mean conceals it: resolved by temperature the model runs high at both ends and low in the middle, a factor of six low near 600 K. Four independent searches reach the same place.

Two readings the data cannot separate: the nucleation term may carry the wrong temperature dependence, or the high- $T$  points may not measure the same population.

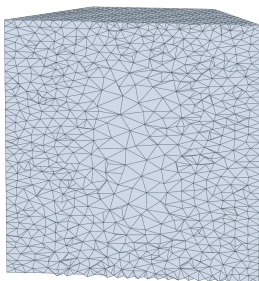
## Four parameters on their bounds

$Z_v^0 = 1.30$  and  $\xi_c = 2.00$  at their ceilings; both coalescence exponents at theirs.

$Z_v^0$  and  $\xi_c$  both multiply the basal capture rate and are nearly degenerate — one statement, not two: **given a stated anisotropy the model cannot deliver the measured basal population without a prefactor no transport argument supplies.**

That is sharper than earlier calibrations supported. With  $p^v$  free it absorbed the discrepancy; pinned, the shortfall is forced into the capture prefactor and *not* the diffusion tensor. It is testable by character-resolved measurements at 600–700 K.

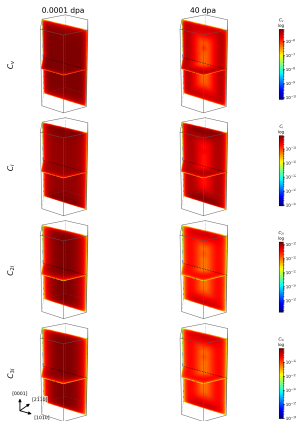
## The demonstration case



|                       |                              |
|-----------------------|------------------------------|
| Geometry              | hexagonal prism              |
| Width $\times$ height | 500 $\times$ 800 nm          |
| Elements              | 60 965 tets                  |
| CD nodes              | 90 663                       |
| Boundary layer        | 100 nm at 20 nm              |
| Temperature           | 573 K                        |
| Dose rate             | $10^{-7}$ dpa s $^{-1}$      |
| Applied stress        | zero                         |
| Dose range            | $10^{-4} \rightarrow 40$ dpa |
| Intervals             | 13                           |

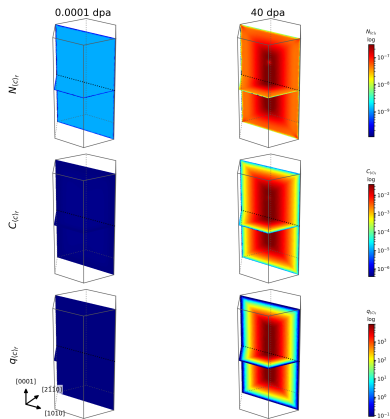
Dirichlet on every face, so the entire surface is grain boundary. The mesh is cut in half so the interior refinement is visible.

# Mobile species, early and late



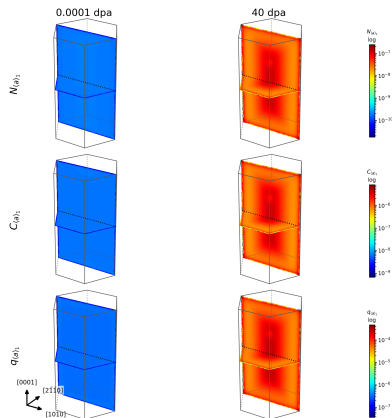
$10^{-4}$  dpa (left) and 40 dpa (right), on two orthogonal mid-cuts. The boundary layers order themselves by species mobility — the vacancy, least mobile, is depleted over the shortest distance — and narrow by a factor of two as  $k^2$  grows with the loop population the same calculation is building.

# The three moments: basal faulted family $c_f$



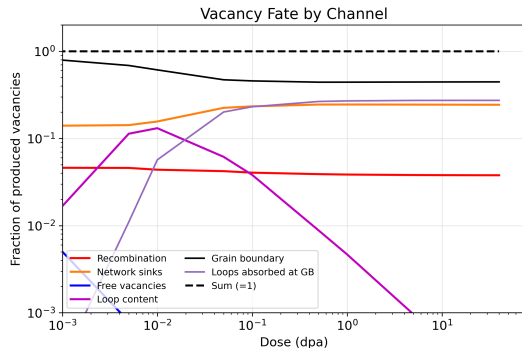
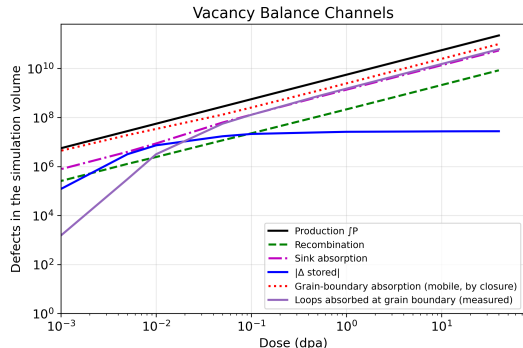
Number density, content and second moment at  $10^{-4}$  and 40 dpa. The three rows are not three views of one thing — they answer different questions and their spatial structures differ, which is the argument for carrying all three.

## The three moments: prismatic variant $a_1$



The same three moments for one prismatic interstitial variant. The prismatic family stays small and numerous where the basal family grows large and sparse — the two are not scaled versions of one another.

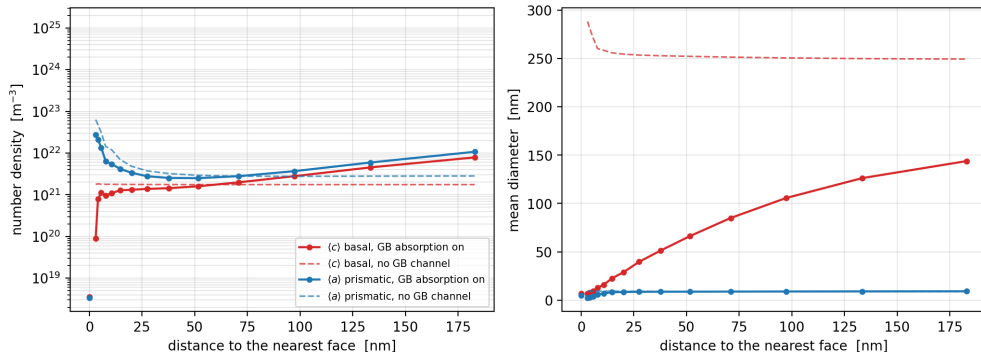
# The atom balance closes — on an open system



The six accumulators integrate a 0-D balance with *no transport term*, so on a Dirichlet domain the residual is not solver error — it is the atoms that left through the surface. At 40 dpa the boundary absorbs **96%** of interstitial production as mobile species and a further **27%** of vacancy production as loops.

# The loop-denuded zone

loop-denuded zone at 40 dpa — 20260831\_072855\_d4fbc06\_sec7\_calibrated\_40dpa



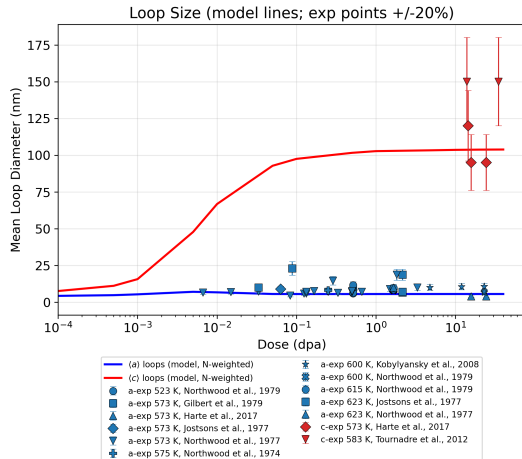
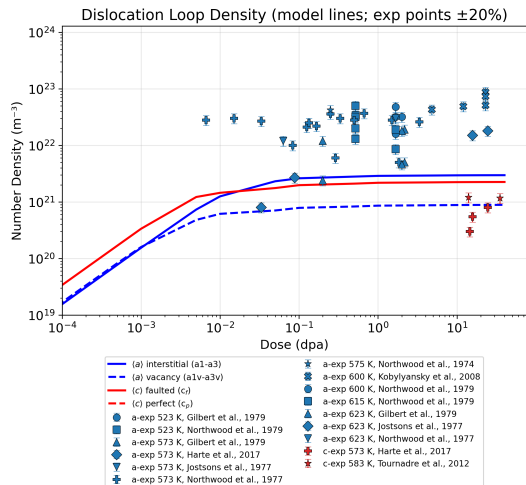
Solid: grain-boundary loop absorption active. Dashed: the identical calculation without it, differing in that one switch and nothing else. The zone is  $\sim 5$  nm wide, and across it the mean loop diameter recovers from **6.8** to **144 nm**. The channel also *raises* the  $\langle c \rangle$  density in the deep interior by a factor of 4–5, because removing the shell's sinks changes the mobile field everywhere — something a mean-field  $k_{gb}^2$  structurally cannot do.

## Domain mean or interior mean? It is not a detail

|                  | Region   | $N_a$ ( $\text{m}^{-3}$ ) | $d_a$ (nm) | $N_c$ ( $\text{m}^{-3}$ ) | $d_c$ (nm) |
|------------------|----------|---------------------------|------------|---------------------------|------------|
| loop channel off | domain   | $3.79 \times 10^{23}$     | 4.29       | $3.98 \times 10^{23}$     | 17.60      |
|                  | interior | $2.80 \times 10^{21}$     | 5.79       | $1.75 \times 10^{21}$     | 249.84     |
| loop channel on  | domain   | $3.86 \times 10^{21}$     | 5.58       | $2.26 \times 10^{21}$     | 103.79     |
|                  | interior | $7.06 \times 10^{21}$     | 5.98       | $5.31 \times 10^{21}$     | 133.01     |

At 40 dpa. **Without** the channel the two differ by  $135\times$  in  $N_a$  and  $227\times$  in  $N_c$  — the domain average is measuring the boundary shell and nothing else. **With** it the disparity collapses to  $1.8\times$  and  $2.4\times$  and changes sign, the interior now carrying more loops than the domain mean. That is the strongest single argument for the channel: it removes an artifact that makes the most natural summary statistic meaningless.

# Loop density and size against dose



Model lines against the measured points,  $10^{-4} \rightarrow 40$  dpa. The march now reaches beyond the  $\langle c \rangle$  measurements at 35 dpa, so the comparison is at matched dose rather than extrapolated. Everything saturates by  $\sim 1$  dpa: from 1 to 40 dpa the densities move under 9% and the radii not at all.

## Against experiment — and the direction of the discrepancy

At 573 K and 24.5 dpa the measurements give  $N_c = 8.0 \times 10^{20} \text{ m}^{-3}$  and  $d_c = 95 \text{ nm}$ . The interior at 40 dpa:

|                  | $N_c/N^{\text{exp}}$ | $d_c/d^{\text{exp}}$ |
|------------------|----------------------|----------------------|
| loop channel on  | 6.6                  | 1.40                 |
| loop channel off | 2.2                  | 2.63                 |

### The direction has changed with the refit

Earlier parameter sets produced *more, smaller*  $\langle c \rangle$  loops — the signature of a model removing loops faster than it grows them. This one produces **more and larger**: stored basal content, which scales as  $N_c d_c^3$ , is 18× what is observed.

The two errors no longer oppose each other, and that is a harder failure to excuse. It is not a size problem a growth term could absorb — it is too much vacancy content retained in basal loops.

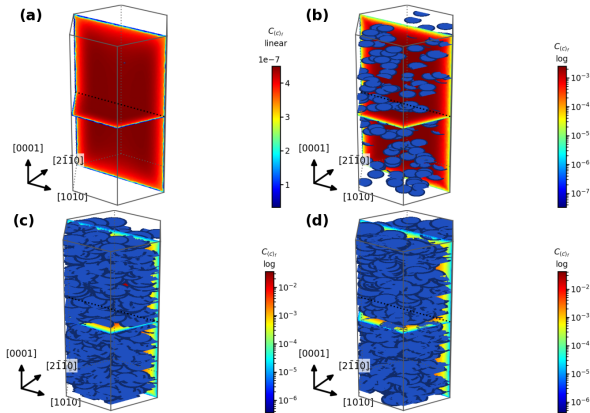
## The conversion, exercised at four doses

| Family                          | $10^{-4}$ dpa | $10^{-2}$ dpa | 1 dpa | 40 dpa |
|---------------------------------|---------------|---------------|-------|--------|
| $c_f$ (basal, vacancy)          | 5             | 238           | 642   | 691    |
| $a_1$ (prismatic, interstitial) | 1             | 55            | 213   | 232    |
| $a_1^v$ (prismatic, vacancy)    | 1             | 23            | 69    | 75     |

The number of loops is not a parameter of the conversion: it is  $\sum_j n_j V_j$ , the number the field says is present. The ledger closes on stored defects to machine precision and on loop number to the integer draw at every one of the four doses.

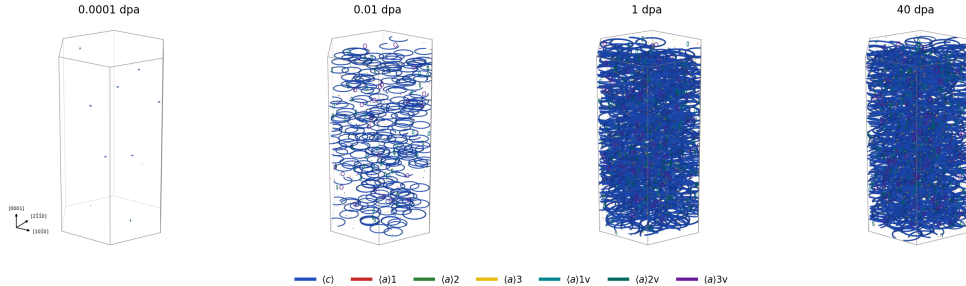
The three interstitial prismatic variants stay *exactly* equal in number (232 each at 40 dpa), which is the isotropic variant weighting under zero applied stress; the three vacancy variants likewise (75 each).

# The discrete basal population



(a)  $10^{-4}$ , (b)  $10^{-2}$ , (c) 1 and (d) 40 dpa, drawn at true radius over the continuum content field. Loops are clipped at the crystal surface for drawing; the exported loops are whole.

# Every family superimposed



Basal  $\langle c \rangle$  in blue, the interstitial prismatic variants and the vacancy variants in their own colours, at true radius and sampled position. What the sequence shows is not the basal family becoming more numerous but its loops becoming very much *larger*, until at 40 dpa they dominate the drawing at a diameter an order of magnitude above the prismatic variants.

## Two regimes, and the limits of the mean field

### At $10^{-4}$ dpa

5 basal and 6 prismatic loops in  $1.3 \times 10^8 \text{ nm}^3$  — a population so dilute that a continuum density describing it is a statement about an *ensemble*, not about this crystal.

### At 40 dpa

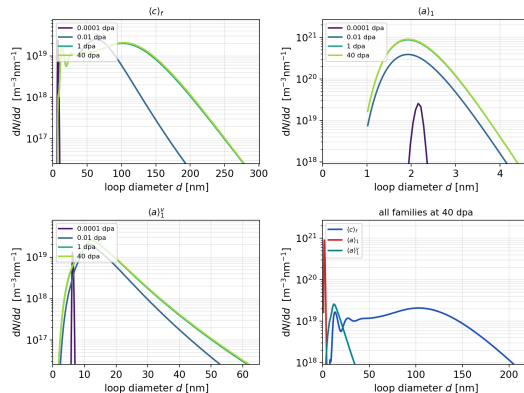
Basal loops 131 nm across at a mean spacing of 57 nm — each loop is *wider than the distance to its neighbour*. This is where the Avrami kernels work hardest and are least defensible.

### And most basal loops do not fit

At 40 dpa, **419 of 691**  $\langle c \rangle$  loops (60.6%) are centered within their own radius of a face and overhang it, by up to 88 nm; the prismatic figure is 3.5%. A discrete-dislocation solver would decline them, and their defects would remain in the continuum.

The conversion spans both regimes, which is what makes it a check on the mean field rather than a convenience — and the refusal fraction is the argument for a larger crystal.

# Reconstructed size distributions



Population per unit  $\ln d$  per cubic metre at four doses, over six decades below the peak. By 1 dpa the *shape* has essentially stopped changing: the 1 dpa and 40 dpa curves are nearly coincident in every panel, so the population saturates in shape as well as in its aggregates.

## One node misbehaves, and it is structural

At 1 and 40 dpa a single node of the 90 663 — node 51 609, at  $x = 214.2$  nm, the point of the prism *furthest* from any face — carries  $\Delta = 4.7 \times 10^3$  and  $6.2 \times 10^3$ , against a median of 1.070. Every other node stays below  $\Delta = 1.5$ .

### The mechanism is in the moment equation

Drift carries  $\Delta_k^{3/8}$  and spreading  $\Delta_k^{-1/8}$ : a family that has already broadened broadens *faster* while its spreading is damped — the dispersion is self-amplifying through the drift. Everywhere else that feedback is held in check by coalescence and by the finite mobile flux; at the one point where the flux is largest and no boundary channel acts, it is not.

$\Delta \geq 1$  is never violated, so no realizability check fires. The remedy is not a tolerance — it is a statement about where the closure is valid, and the gap between the outlier and the rest of the crystal is three orders of magnitude with nothing in between.

## What is established

- Every bias follows from the diffusion tensors through one  $p^m$  per species. The anisotropy is an **input**, not a fitted absorber.
- Spatial resolution matters: boundary layers order by mobility and narrow by  $2\times$ ; a domain mean without the loop channel measures the boundary shell.
- The transfer preserves what defines the population, so continuum / discrete / hybrid is a **cost** decision.
- $2.1 \times 10^8$  stiff equations in 1 h 10 m on 8 cores.

## What is open

- The basal density,  $2.1\times$  the measurement, with the capture prefactor on its bound — absorption, not transport.
- The prismatic temperature trend, a factor of six low near 600 K.
- $p^i = 0.70$  implies 0.106 eV of migration anisotropy at 573 K — **open to atomistic test**.
- Whether  $c_p$  needs a sink of its own; it is a terminal population.

The useful outcome is that a shortfall now has to appear somewhere it can be named. With the anisotropy pinned, the model cannot quietly refit a bias to hide it.